

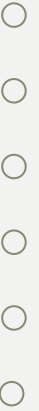
“Touching the Window”

Exploring Conductive Ink as Touch Input

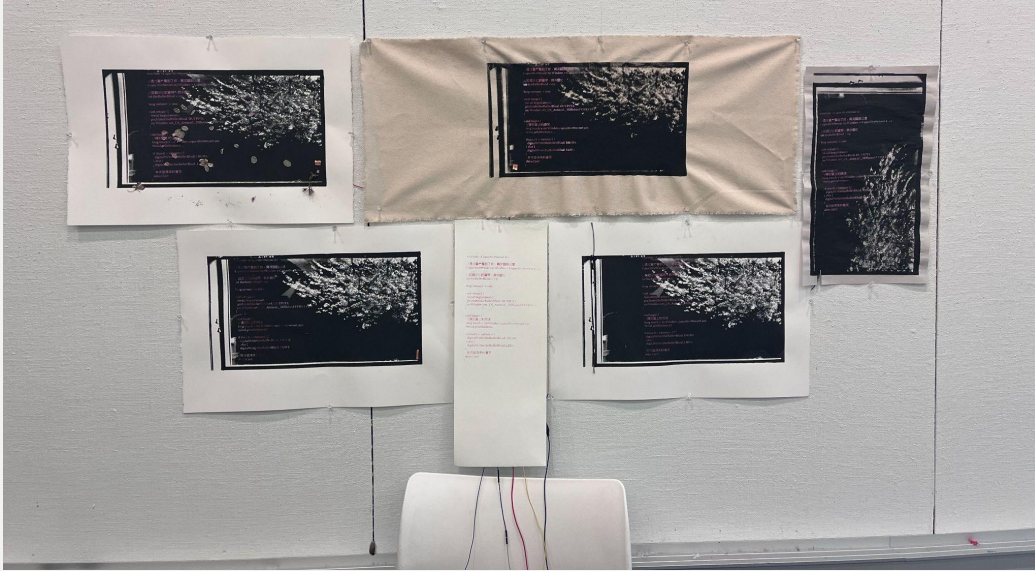
Amy Yiqin Li



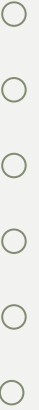
Video



What is it..



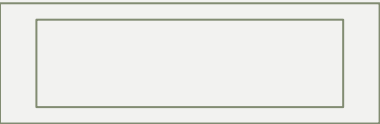
An interactive installation where touch activates sound through printed conductive surfaces.



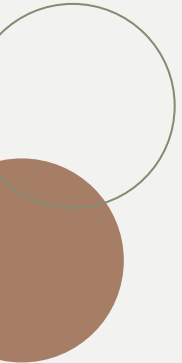
How does it work?



- DIY conductive ink
- Turn the print into a capacitive sensor that sense electrical capacitance of human body
- Connected to arduino and touchdesigner to control audio



Making the ink



Solvent

Water or glycerin.
Controls viscosity

Binder

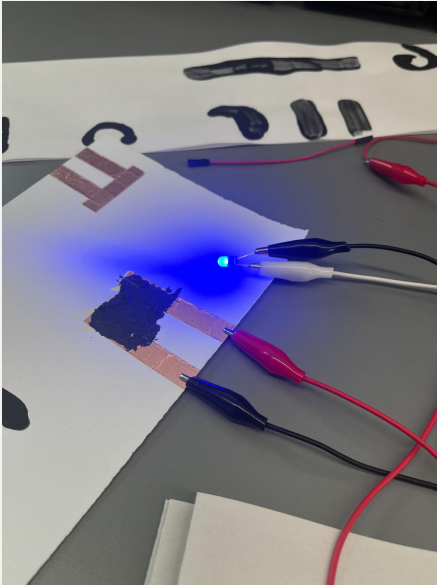
Water-based screen
print ink/ White glue
Make the ink stays on
the surface of the
object

Conductive Particles

graphite powder



Experimentation & Iteration

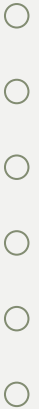


Ink Test

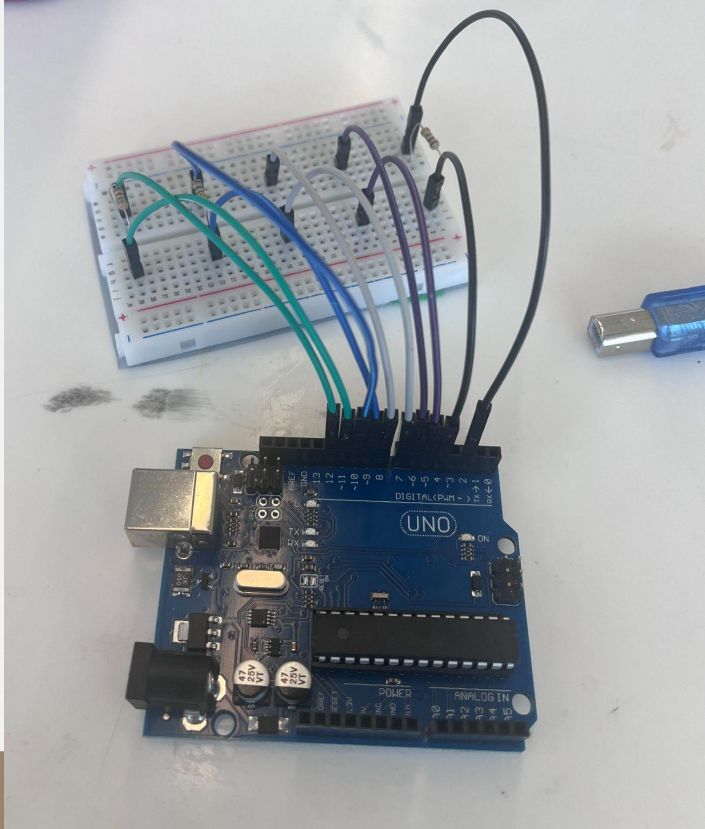
Screenprint
Test



Printed Images



circuits



A send pin that connects to the
conductive surface,

A receiver pin

A medium to high value resistor.

(I used 1M)

Project Setup

01 The Prints

03 Arduino + Touchdesigner

Processing data and control audio



02 Cables

connecting with the print through copper tape

04 Speaker

Output for audio

Touchdesigner

Controls multi-channel
audio

Can calibrate the data in real
time

Easy to connect with Arduino



Future Improvements

- Better signifier
- Human touch makes the image blurry
- Needs a lot of recalibration
- Better audio(?)



References

- CapacitiveSensor - Arduino Library
- Soft Inkjet Circuits: Rapid Multi-Material Fabrication of Soft Circuits
- An introduction to printed electronics
- Screen-printed electrodes
- Conductive inks for printed electronics
- PaperID: A Technique for Drawing Functional Battery-Free Wireless Interfaces on Paper
- BUILDING a Capacitive Sensor with ARDUINO

